



QR Tool Management System

Test Case Report

Prepared on: 04 July 2026

This report documents the test cases covering the QR Tool Management System's backend API, database layer, and frontend application. Backend and Database sections were executed as an automated pytest suite (backend/tests/) against an isolated in-memory database and are reported with a real Pass/Confirmed status. Frontend sections are documented functional test cases based on a detailed review of the application's actual behavior; as no automated frontend test harness is configured in this project yet (see Known Gaps), they are marked Pending Execution rather than Passed. Two real defects found while writing these test cases were fixed directly in this pass rather than only being logged - see Section 4, Fixes Applied.

Summary

Area	Test Cases	Result
Backend / API (automated, pytest)	52	All Passed
Database (constraints & integrity)	10	Passed / Confirmed by inspection
Frontend (documented functional cases)	71	Pending manual/automated execution
Known gaps identified	4	See Known Gaps section
Defects fixed this pass	3	See Fixes Applied section

1. Backend / API Test Cases (Automated)

Executed via **python -m pytest backend/tests** against an isolated in-memory SQLite database, with the app's real dependency-injection wired through a test client. All 52 tests passed.

Authentication & Registration

ID	Test Case	Expected Result	Status
AUTH-01	Create user without a verified email	Rejected with 400 - email has not been verified	Passed
AUTH-02	Create user after email OTP is verified	User created successfully (200), password hash never returned	Passed
AUTH-03	Create user with a username already registered	Rejected with 400 - username already registered	Passed
AUTH-04	Create user with a phone number already registered to another account	Rejected with 400 - mobile number already registered	Passed
AUTH-05	Login as a Worker-role user	Access token returned directly, no OTP step required	Passed
AUTH-06	Login as a non-Worker role (e.g. Store)	Response has otp_required=true with the masked destination email	Passed
AUTH-07	Complete the OTP flow for a non-Worker role	Correct OTP exchanges for a valid access token	Passed
AUTH-08	Submit an incorrect OTP code	Rejected with 400 - invalid or expired verification code	Passed
AUTH-09	Login with an incorrect password	Rejected with 401 Unauthorized	Passed
AUTH-10	Login with an unknown username	Rejected with 401 Unauthorized	Passed
AUTH-11	Login to a deactivated (status=inactive) account	Rejected with 403 - account deactivated	Passed
AUTH-12	Login using the registered email instead of username	Login succeeds identically to username login	Passed

Users & Access Control

ID	Test Case	Expected Result	Status
USR-01	List users with no auth token	Rejected with 401 Unauthorized	Passed
USR-02	List users with a valid token	200 OK, returns a list of users	Passed
USR-03	GET /users/me	Returns the authenticated user's own profile	Passed
USR-04	User updates their own profile	200 OK, field changes are persisted	Passed
USR-05	Non-admin user updates a different user's profile	Rejected with 403 Forbidden	Passed
USR-06	Admin updates another user's profile	200 OK, field changes are persisted	Passed
USR-07	Update a user ID that does not exist	404 Not Found	Passed
USR-08	Attempt to delete a user (any role, any user ID)	Rejected with 400 - deletion is permanently disabled in favor of the disable toggle	Passed

ID	Test Case	Expected Result	Status
USR-09	Admin disables a user via PATCH status=inactive	200 OK; that user is subsequently blocked from logging in (403)	Passed
USR-10	Attempt to delete a user ID that does not exist	Rejected with 400 - same as any other delete attempt	Passed

Dealers

ID	Test Case	Expected Result	Status
DLR-01	Create a dealer with no auth token	Rejected with 401 Unauthorized	Passed
DLR-02	Worker role creates a dealer	Rejected with 403 - not authorized to manage dealers	Passed
DLR-03	Store role creates a dealer	200 OK, dealer created	Passed
DLR-04	Admin role creates a dealer	200 OK, dealer created	Passed
DLR-05	Create a dealer with an invalid category	Rejected with 400 - invalid dealer category	Passed
DLR-06	Create a dealer with a dealer_code that already exists	Rejected with 400 - dealer code already exists	Passed
DLR-07	Create a dealer with a lower-case dealer_code	dealer_code is normalized to upper-case on save	Passed
DLR-08	List dealers filtered by category	Only dealers in the requested category are returned	Passed

Inspections

ID	Test Case	Expected Result	Status
INSP-01	Create an inspection with no auth token	Rejected with 401 Unauthorized	Passed
INSP-02	Submit an inspection with result=usable	Tool status becomes usable, inspection_result=usable	Passed
INSP-03	Submit an inspection with result=fail	Tool status becomes scrap, inspection_result=not-usable	Passed
INSP-04	Submit an inspection with result=repair	Tool status becomes under-repair	Passed
INSP-05	Submit an inspection with usability_percentage=50	A critical low-usability Alert is generated for the tool	Passed
INSP-06	Submit an inspection with usability_percentage=95	No low-usability Alert is generated	Passed
INSP-07	Inspector role without a verified employee profile submits an inspection	Rejected with 403 - profile must be verified first	Passed

Tools (Tool Master)

ID	Test Case	Expected Result	Status
TOOL-01	Create a tool as an authenticated user	200 OK, tool created with status=usable	Passed
TOOL-02	Create a tool with no auth token	Rejected with 401 Unauthorized	Passed
TOOL-03	Create a tool with a QR code that already exists	Rejected with 400 - a tool with this QR code already exists	Passed
TOOL-04	Request a tool by an ID that does not exist	404 Not Found	Passed

ID	Test Case	Expected Result	Status
TOOL-05	Retrieve a tool by its ID	200 OK, correct tool data returned	Passed
TOOL-06	List tools after deleting one	Deleted tool is excluded from the default list	Passed
TOOL-07	Search tools by description text	Matching tool(s) are returned	Passed
TOOL-08	Change a tool's current_site via PATCH	A Movement History record (from -> to) is created	Passed
TOOL-09	Update a tool ID that does not exist	404 Not Found	Passed
TOOL-10	Delete a tool that was never printed	Hard delete - the row is fully removed	Passed
TOOL-11	Delete a tool that has been marked as printed	Soft delete (is_deleted=true); admin can restore it afterward	Passed
TOOL-12	Non-admin attempts to restore a deleted tool	Rejected with 403 Forbidden	Passed
TOOL-13	Look up a tool by its full QR code	200 OK, correct tool returned	Passed
TOOL-14	Look up a tool by only the last 4 digits of its QR code	200 OK, correct tool returned	Passed
TOOL-15	Look up a QR code that does not exist	404 Not Found	Passed

2. Database Test Cases

Constraint and integrity checks against the MySQL/SQLModel schema (qr_tools_db). Items already exercised by the API suite above are marked Passed; items verified by direct schema or code inspection (not currently expressed as an isolated automated test) are marked Confirmed.

ID	Test Case	Expected Result	Status
DB-01	user.username has a unique constraint	Duplicate usernames are rejected (see AUTH-03)	Passed
DB-02	tool.qr_code has a unique index	Duplicate QR codes are rejected with a clean 400 before hitting the DB constraint (see TOOL-03)	Passed
DB-03	dealer.dealer_code has a unique constraint	Duplicate dealer codes are rejected (see DLR-06)	Passed
DB-04	Passwords are never stored or returned as plaintext	hashed_password (bcrypt) is excluded from every API response model	Passed
DB-05	inspection.tool_id / inspection.inspector_id are foreign keys	Production MySQL schema enforces FK constraints to tool.id / user.id	Confirmed
DB-06	tool.created_by_id is a foreign key to user.id	Automatically populated from the authenticated user on create	Passed
DB-07	Soft-delete (is_deleted) hides printed tools without losing data	Row and history are preserved; tool disappears from active views (see TOOL-11)	Passed
DB-08	Hard-deleting an unprinted tool cascades correctly	Its Inspection/MovementHistory rows are deleted; related Alert rows have tool_id nulled instead of being deleted	Confirmed
DB-09	Audit log 30-day retention purge	AuditLog entries older than 30 days are deleted every time a new action is logged	Confirmed
DB-10	Live MySQL schema (qr_tools_db) matches the SQLModel schema	All 11 tables match, including company_name/gst_number columns added to `user` during this engagement	Confirmed

3. Frontend Test Cases

Documented functional test cases derived from a detailed review of each page's actual implemented behavior. No automated frontend test harness is configured in this project (see Known Gaps), so these are marked Pending Execution; cases marked Known Gap describe UI actions that do not do what a user would reasonably expect.

Login & Authentication

ID	Test Case	Expected Result	Status
FE-LOGIN-01	Select the "Worker" role card	Logs in immediately as a guest, no credentials form shown, navigates to /worker	Pending Exec.
FE-LOGIN-02	Submit valid credentials for an OTP-required role (e.g. Store)	Transitions to the OTP-verification screen showing the masked email/phone	Pending Exec.
FE-LOGIN-03	Enter the correct OTP	Login completes and routes to that role's landing page (Store -> /store-view)	Pending Exec.
FE-LOGIN-04	Log in with valid credentials under the wrong role card (e.g. a Store user selects Admin)	Rejected client-side with "Invalid username or password"	Pending Exec.
FE-LOGIN-05	An Admin account logs in through any role portal	Login succeeds regardless of the selected role card (admin bypass)	Pending Exec.
FE-LOGIN-06	Upload a QR code image on the login screen before logging in	Scanned code carries through to the destination page after login	Pending Exec.
FE-LOGIN-07	Attempt login while the backend is unreachable	Distinct network/server error message, not the generic invalid-credentials message	Pending Exec.

Dashboard

ID	Test Case	Expected Result	Status
FE-DASH-01	Load the Dashboard	Stat cards (Total, Usable, Scrap, Under Repair, Pending, Expiring, Overdue) reflect current inventory	Pending Exec.
FE-DASH-02	Click the "Deleted (Printed)" stat card	Opens a modal listing soft-deleted tools	Pending Exec.
FE-DASH-03	View Dashboard as a non-admin role	"Filter Creator"/"Filter Store" dropdowns and the Restore action are hidden	Pending Exec.
FE-DASH-04	Admin restores a tool from the Deleted Tools modal	Tool reappears in the active inventory afterward	Pending Exec.
FE-DASH-05	Load Dashboard with unread expiry alerts pending	Alert modal auto-opens; "Acknowledge All" marks them read	Pending Exec.
FE-DASH-06	Apply a filter, then Export PDF/Excel	Exported report reflects the filtered data set, not the full inventory	Pending Exec.

Tool Master

ID	Test Case	Expected Result	Status
FE-TM-01	Submit the New Tool form missing a required field (e.g. Capacity)	Save is blocked; the missing field is highlighted	Pending Exec.
FE-TM-02	Fill in Name/Metal/Variant/Capacity/Date/Purchaser for a new tool	The QR/Tool ID preview updates live as each field changes	Pending Exec.
FE-TM-03	Save a new tool with Inspection Result = "not-usable"	Tool status is set to "scrap" automatically	Pending Exec.

ID	Test Case	Expected Result	Status
FE-TM-04	Upload an existing tool's QR image via "Scan to Autofill"	Form is populated for editing by looking the tool up via its QR code	Pending Exec.
FE-TM-05	Admin/Data-Entry runs Bulk Import with an Excel file	After import, "Zip QR download" and "Excel with QR link" actions appear	Pending Exec.
FE-TM-06	View the Existing Inventory tab as a non-admin role	Edit and Delete actions are not visible	Pending Exec.
FE-TM-07	Click Delete on a tool row	A native browser confirm dialog appears before the delete request is sent	Pending Exec.
FE-TM-08	Type in the search box after navigating to page 2 of the inventory	Pagination resets back to page 1	Pending Exec.

Store Inventory & Store View

ID	Test Case	Expected Result	Status
FE-STORE-01	Open Store Inventory as a Store Manager	Only tools currently at that manager's own site are listed	Pending Exec.
FE-STORE-02	Open Store Inventory with no tools in "pending" status	The Pending Returns panel is not shown	Pending Exec.
FE-STORE-03	Resolve Status -> Extend without entering a reason	Submission is blocked until a reason and a positive day count are entered	Pending Exec.
FE-STORE-04	Resolve Status -> Missing	Tool is permanently marked as missing	Pending Exec.
FE-STORE-05	Select rows and click "Receipt (IN)"	Navigates to Tools Movements with the selection carried in route state	Pending Exec.
FE-STORE-06	Attempt an OUT transaction on a tool that is not "usable"	Blocked with a toast, except Scrap Disposal (requires status=scrap)	Pending Exec.
FE-STORE-07	Attempt "Sub-Contractor Return" on a tool with no subcontractor_name recorded	The option is disabled	Pending Exec.
FE-STORE-08	Enter a mobile number that is not exactly 10 digits	Inline validation error is shown; submission is blocked	Pending Exec.
FE-STORE-09	Complete a successful IN/OUT transaction	An editable Delivery Challan PDF preview is generated before download	Pending Exec.

Tools Movements & Movement History

ID	Test Case	Expected Result	Status
FE-MOVE-01	Navigate to Tools Movements with no bulk selection made	Empty-state message directs the user back to Store Inventory	Pending Exec.
FE-MOVE-02	Run a Bulk OUT to a subcontractor/site/store with a mix of tool statuses selected	Only tools currently "usable" are actually processed	Pending Exec.
FE-MOVE-03	Submit Bulk OUT for Sub-Contractor Work without a valid 10-digit mobile number	Submission is blocked with an inline error	Pending Exec.
FE-MOVE-04	Search Movement History by tool/site/remarks/user	Results are scoped to the logged-in store's own site and match the search text	Pending Exec.

Inspection Results & Inspection Employees

ID	Test Case	Expected Result	Status
FE-INSP-01	Open Inspection Results	Page is read-only - no create/edit controls are present	Pending Exec.
FE-INSP-02	Search Inspection Results	Filters across tool description, QR code, result, remarks, and inspector name	Pending Exec.
FE-INSP-03	Admin clicks "Verify" on a Pending inspection employee	Badge changes to Verified	Pending Exec.
FE-INSP-04	Admin clicks "Unverify" on a Verified inspection employee	Badge reverts to Pending	Pending Exec.
FE-INSP-05	Click an inspection employee's name	Opens a dialog listing every tool that employee has inspected	Pending Exec.

Dealers

ID	Test Case	Expected Result	Status
FE-DEALE R-01	Switch between Sub-Contractor / Supplier / Scrap-Dealer tabs	Field labels change and the Products/Services dropdown resets	Pending Exec.
FE-DEALE R-02	Submit Add Dealer without Name, Company Name, or Dealer Code	Submission is blocked	Pending Exec.
FE-DEALE R-03	Enter a lower-case Dealer Code and submit	Code is uppercased automatically	Pending Exec.
FE-DEALE R-04	Click Delete on a dealer	Native confirm dialog appears before the delete request is sent	Pending Exec.
FE-DEALE R-05	View the page as a non-admin (Store) user	Bulk Import and Custom Fields management are not visible	Pending Exec.
FE-DEALE R-06	Search/paginate one category tab (e.g. Suppliers)	Other category tabs' search/pagination state is unaffected	Pending Exec.

Users Management

ID	Test Case	Expected Result	Status
FE-USERS -01	Attempt to add a user before verifying the email OTP	"Add User" is blocked until Send OTP -> Verify succeeds	Pending Exec.
FE-USERS -02	Change the email after OTP verification succeeded	Verified flag resets, forcing re-verification before the user can be added	Pending Exec.
FE-USERS -03	Select Role = Store	Site field switches to free text for creating a brand-new store location	Pending Exec.
FE-USERS -04	Select Role = Management	Site field is hidden (not required)	Pending Exec.
FE-USERS -05	Enter a phone number that fails the 10-digit format check	Inline validation error is shown	Pending Exec.
FE-USERS -06	View the combined user table	Regular users and Inspection Employees appear together, inspector IDs prefixed "insp-"	Pending Exec.
FE-USERS -07	Add a new user with Role = Store and a new site name	New site is immediately available in other Site dropdowns without a page reload	Pending Exec.

Alerts & Audit Log

ID	Test Case	Expected Result	Status
FE-ALERT-01	Open Alerts and switch between Critical/Warning/Info/All tabs	Unread counts per severity match the tab contents	Pending Exec.
FE-ALERT-02	Click "Mark as Read" on a single unread alert	Alert state updates via the read endpoint	Pending Exec.
FE-ALERT-03	Click "Mark All as Read"	Every unread alert is marked read via the API (fixed - previously only showed a toast with no API call)	Pending Exec.
FE-ALERT-04	Search the Audit Log	Filters across description, username, action, entity_type, and site	Pending Exec.
FE-ALERT-05	Export PDF/DOC from Audit Log with an empty filtered result set	Export buttons are disabled; when enabled, they export only the filtered rows	Pending Exec.

Reports

ID	Test Case	Expected Result	Status
FE-REPORT-01	Combine the global search box with per-column filters, then click "Clear Filters"	Both the global search and column filters are reset together	Pending Exec.
FE-REPORT-02	Apply a filter, then Export PDF/CSV	Export reflects only the filtered result set	Pending Exec.
FE-REPORT-03	View a tool row that has a test_certificate on file	A working certificate download link is shown	Pending Exec.
FE-REPORT-04	Click a tool's description	Opens a detail card with QR code, supplier info, and site-movement history	Pending Exec.

Settings & Tool Config

ID	Test Case	Expected Result	Status
FE-SET-01	Enter an invalid Company Email on the General tab and click Save	Save is blocked with an inline format error	Pending Exec.
FE-SET-02	Enter a Phone number on the General tab in an invalid format	Inline validation error is shown	Pending Exec.
FE-SET-03	Change General/Notifications/Security settings, save, then refresh the page	KNOWN GAP: changes revert - there is no backend persistence for these tabs	Known Gap
FE-SET-04	Toggle a role under Dashboard Access Permissions (Security tab)	Takes effect immediately for that role via localStorage - the one functionally real Settings feature	Pending Exec.
FE-SET-05	Click "Export Reports" on the Data & Backup tab	KNOWN GAP: no handler is wired - the button does nothing	Known Gap
FE-SET-06	Click "Create Backup Now"	Backup PDF is emailed to the registered address; success toast shows that address	Pending Exec.
FE-SET-07	Delete a Tool Config entry that has already-printed matching tools	Deletion is blocked with an explanatory message instead of failing silently	Pending Exec.

QR Scanner

ID	Test Case	Expected Result	Status
FE-QR-01	Click "Start Camera Scanner"	Requests camera permission; shows an inline error if denied/unavailable	Pending Exec.

ID	Test Case	Expected Result	Status
FE-QR-02	Successfully scan a valid QR code	Scanner Dialog auto-closes and passes the extracted code (URL prefix stripped if present) to the parent page	Pending Exec.
FE-QR-03	Point the camera away from any QR code briefly	Routine "no QR in frame" exceptions are silently ignored, not shown as errors	Pending Exec.

4. Fixes Applied This Pass

Real defects surfaced by writing and running these test cases were fixed directly, then re-verified by re-running the full automated suite (52/52 passing) before this report was regenerated.

- TOOL-03 / DB-02: creating a Tool with a duplicate QR code previously surfaced as an unhandled 500 error; backend/routes/tools.py now checks for an existing qr_code first and returns a clean 400 Bad Request.
- FE-ALERT-03: the Alerts page's "Mark All as Read" button previously only showed a success toast with no API call; it now calls POST /alerts/{id}/read for every unread alert and updates state from the result.
- The Alerts page's dead, never-wired handleResolve function (no button actually triggered it) was removed as unused code rather than left as a misleading stub.

5. Known Gaps

- No automated frontend test harness (Vitest/React Testing Library or Playwright) is configured yet - all Frontend test cases in this report are documented for manual execution rather than run in a browser.
- Settings General/Notifications/Security tabs do not persist to the backend - only the Dashboard Access Permissions toggle actually takes effect (FE-SET-03/04).
- Settings → Data & Backup → "Export Reports" button has no handler wired (FE-SET-05).
- The legacy backend/test_api.py script predates the /api routing prefix used throughout main.py and no longer runs against the current app; it has been superseded by the backend/tests/ pytest suite.

6. Recommendations

- Add Vitest + React Testing Library (component/unit level) and/or Playwright (end-to-end) so the Frontend test cases in this report can run automatically in CI rather than remaining manual.
- Either implement or remove the stubbed Settings actions (General/Notifications/Security persistence, "Export Reports") so the UI does not imply a persisted change that never happens.
- Retire or update backend/test_api.py to match the current /api-prefixed routes, or delete it in favor of the backend/tests/ suite to avoid confusing future contributors.